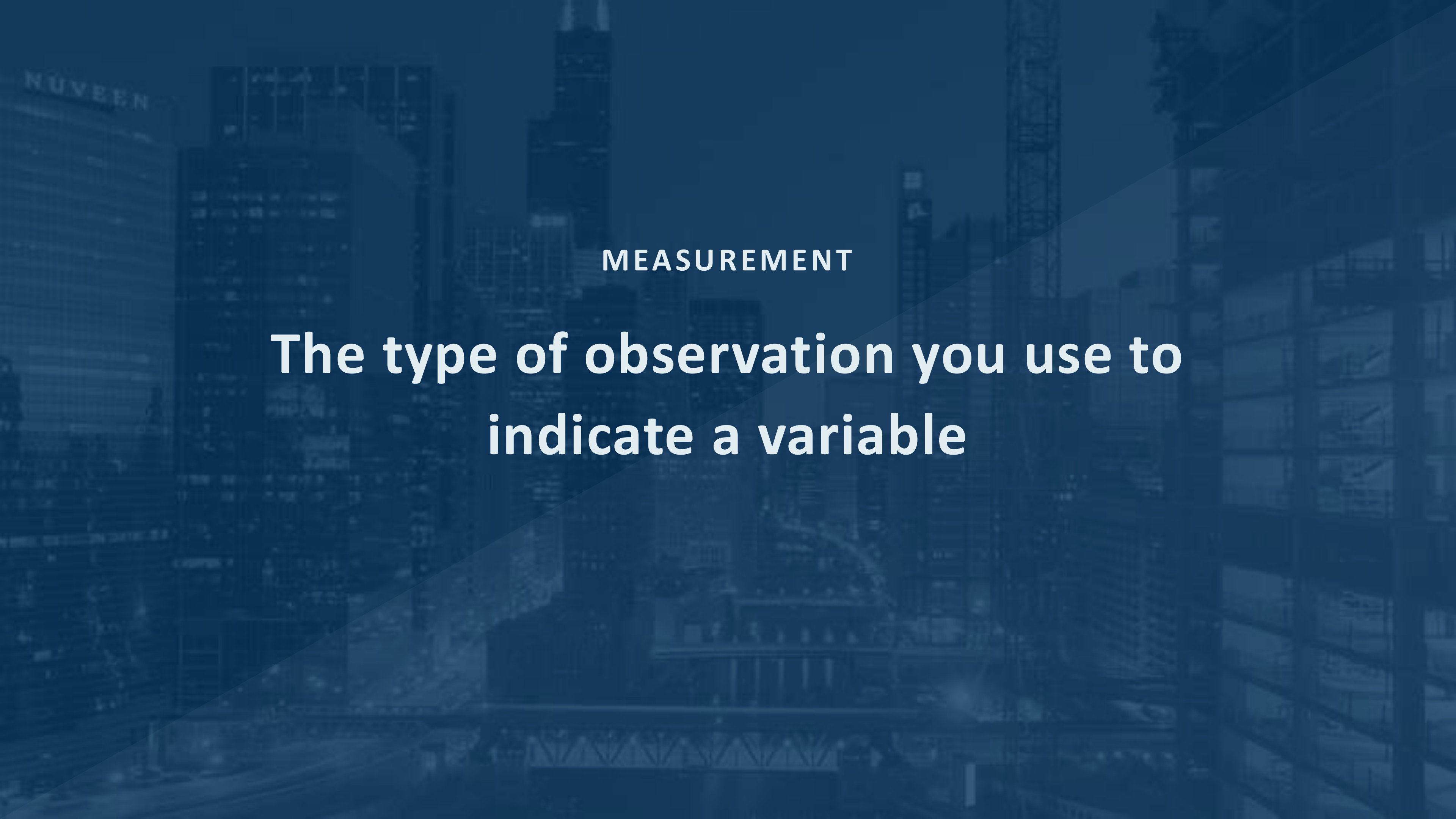


CJ 540

MEASUREMENT

Professor Burrell Vann Jr



MEASUREMENT

**The type of observation you use to
indicate a variable**

MEASUREMENT

How would you measure?

- political party affiliation
- age
- grade point average
- religious affiliation



MEASUREMENT

HOW WOULD YOU MEASURE?

- prejudice
- compassion

MEASURING THE SOCIAL WORLD WITH CONSTRUCTS

CONSTRUCTS

Theoretical variables, not observed directly or indirectly but having agreed upon meanings (e.g. intelligence)

Have no intrinsic reality, only the meaning we agree to give it

CLEAR

AGE
RACE/ETHNICITY
HEIGHT
RELIGIOUS AFFILIATION

CONSTRU

CT
RACISM
THREAT
FEAR
SATISFACTION



OPERATIONALIZ ATION

Defining how a variable (construct or clear) will
be measured

OPERATIONALIZING CONSTRUCTS

CONSTRUCT

Theoretical, difficult to measure

- Racism

DIMENSION

a particular aspect of a construct

- Institutional Racism

INDICATOR

A direct observation we choose to reflect (a dimension of) a construct

- Job rejection
- Loan denial
- Segregated housing

OPERATIONALIZING CONSTRUCTS





OPERATIONALIZING CONSTRUCTS

Defining how a variable (construct or clear) will be measured...

"In this study of Black individuals, I focus on the experience with racism. The first variable, racism, centers specifically on institutional racism. I measure institutional racism using 3 indicators: whether the black individuals in my sample were 1) denied a job, 2) denied a loan, and 3) currently living in segregated housing. All of these measures were coded as either 'Yes' or 'No'."



CLASSIFYING PARTICIPANT RESPONSES

MUST BE EXHAUSTIVE & MUTUALLY EXCLUSIVE

EXHAUSTIVE

Must include options for all possible attributes of a variable to correctly classify all participants.



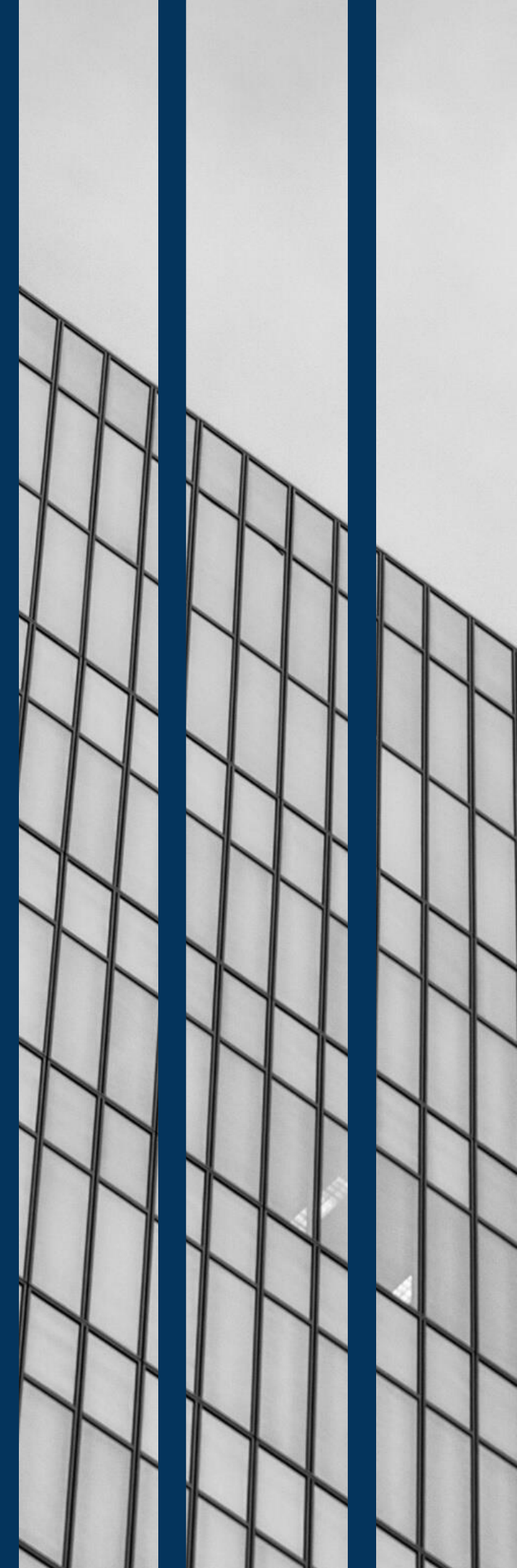
MUTUALLY EXCLUSIVE

With a few exceptions, each participant should only be able to select/be classified as ONE attribute for each variable



Levels of Measurement

- ▶ **NOMINAL**
least sophisticated
- ▶ **ORDINAL**
more sophisticated
- ▶ **INTERVAL-RATIO**
most sophisticated



LEVELS OF MEASUREMENT

NOMINAL ORDINAL

Attributes have only the characteristics of exhaustiveness and mutually exclusiveness. Simple categories.

Attributes that classify, but cannot be ranked

- *religious affiliation, college major, hair color, birthplace, nationality*

Attributes that classify and can be logically rank-ordered but lack an associated numerical value

- *socioeconomic status, level of conflict, prejudice, conservativeness*

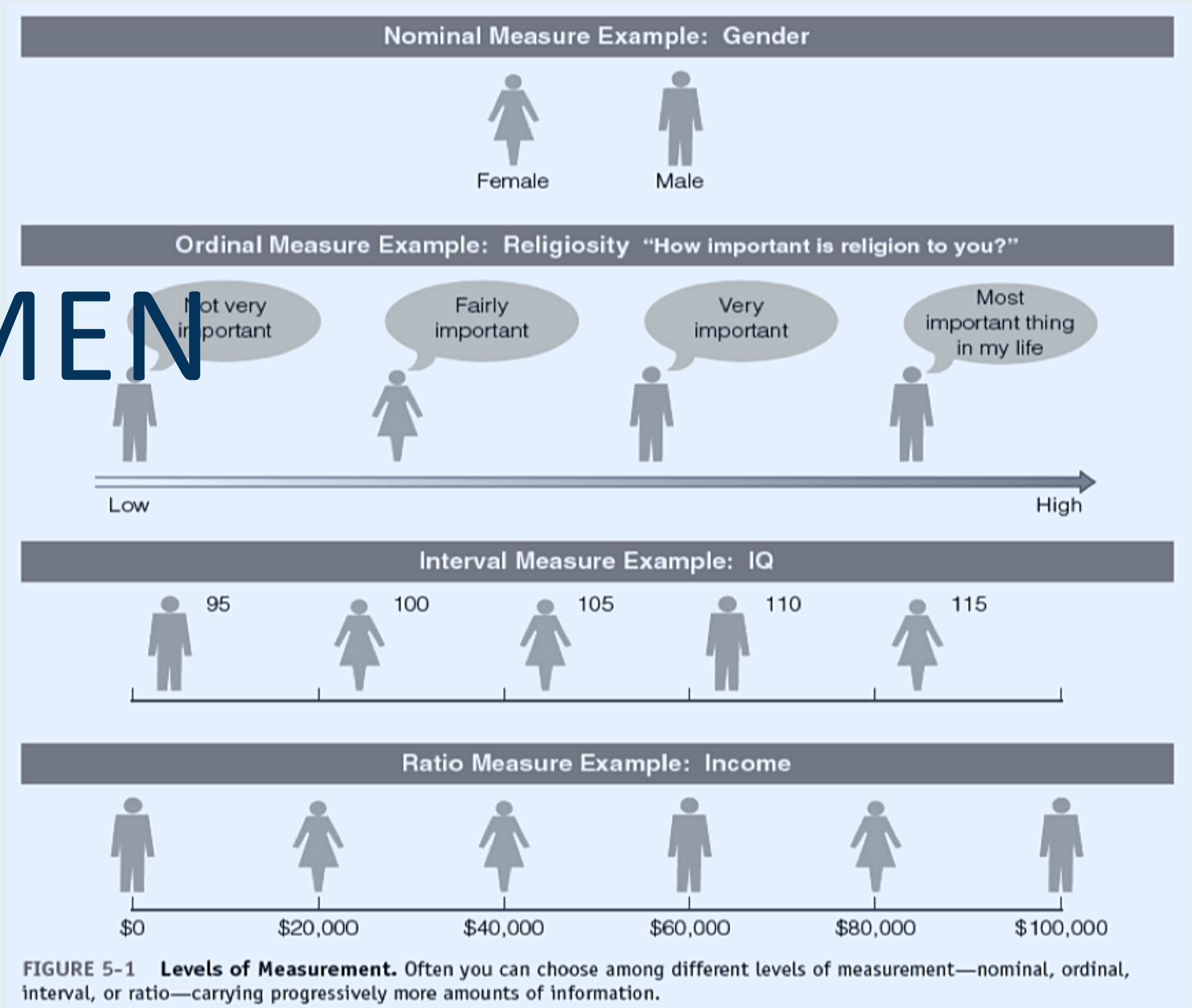
INTERVAL-RATIO

Distance between attributes has meaning.

Attributes classify, can be rank-ordered, and have an equal numerical distance between values

- *temperature (Fahrenheit), IQ score*

LEVELS OF MEASUREMENT



ENSURING RELIABILITY

Y SAME OBSERVATIONS WOULD
OCCUR WITH REPEATED
MEASUREMENTS OF THE SAME
PHENOMENON

▶ **TEST-RETEST**

making the same measurement more than once

▶ **SPLIT-HALF**

multiple sets of randomly assigned variables should produce the same classifications

▶ **ESTABLISHED MEASURES**

using measures that have proven reliable in other research

▶ **INTER-RATER RELIABILITY**

testing how close research workers are in their coding of the data

ENSURING VALIDITY

MEASUREMENTS THAT
ACCURATELY REFLECT THE THING
IT IS INTENDED TO MEASURE

▶ **FACE VALIDITY**

measurement that seems to make reasonable sense

▶ **CRITERION-RELATED**

*measurement related to some external
criterion/outcome (e.g. written DMV test and clean
driving records)*

▶ **CONSTRUCT**

*degree to which a measure relates to other variables
as expected within a system of theoretical relationships
(e.g. marital satisfaction related to marital fidelity)*

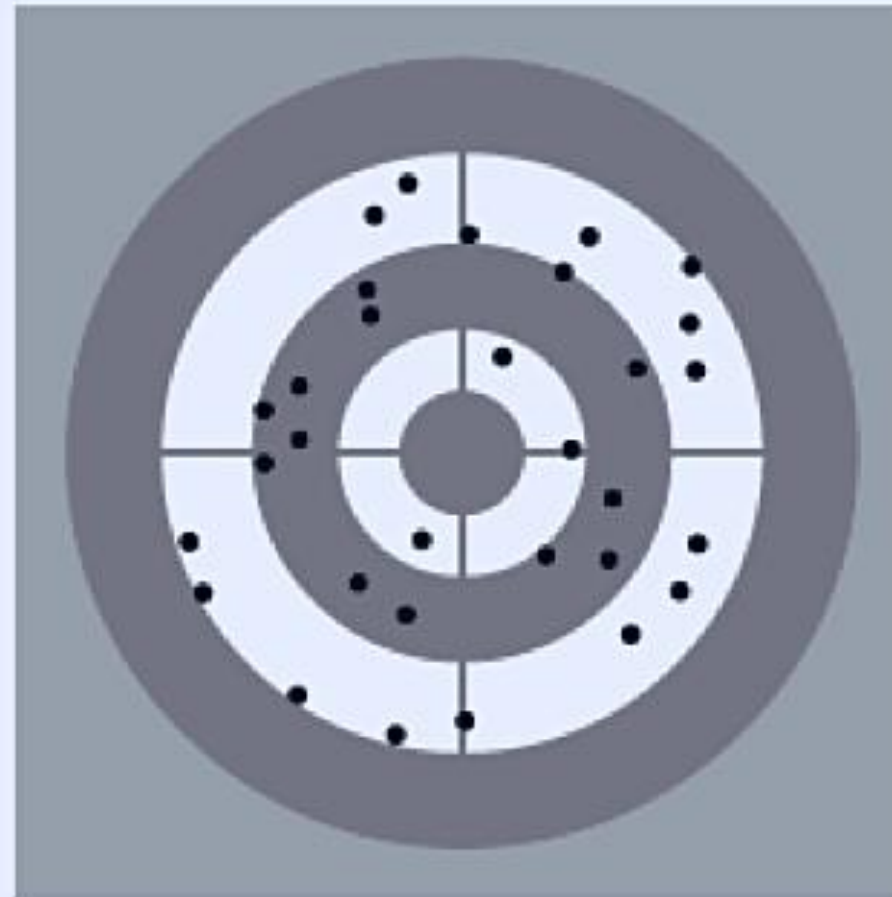
▶ **CONTENT**

*measure that covers the range of meanings included
within a concept (e.g. math ability covers +/- , algebra,
precal, calc)*

NO VALIDITY WITHOUT RELIABILITY



Reliable but not valid



Valid but not reliable



Valid *and* reliable